

Cross-Distribution Identity-Document Fraud Detection: Diversity-Axis Ensembles of Double-Trace Consistency Detectors

The FREUID Challenge 2026: Technical Report (Team ETRI_MobilityAI)

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Abstract

The FREUID Challenge 2026 requires a scalar fraud score in $[0, 1]$ for each identity-document image, ranked by the FREUID Score ($\text{AuDET} + \text{APCER}@1\%\text{BPCER}$; lower is better). The test data has a deliberate distribution shift. The public split (about 5% of the test data) is born-digital. The private split (about 95%) decides the ranking. It is drawn from a hybrid of synthetic and physically captured images, and it includes two document types absent from training. A single submission is scored on both splits, and the private split dominates. The FREUID Score is a strict-tail metric, so it punishes any subpopulation on which a model collapses. We therefore submit one model that is built for robustness to captured and unseen-type content, not the model that tops the born-digital public leaderboard. The submitted model is a Double-Trace Consistency (DTC) capture ensemble. Each DTC detector adds a high-pass forensic branch to a pretrained RGB backbone and uses the trace twice: as a global feature and as a document-type-agnostic patch-consistency signal. The main gain comes from ensembling backbones that are decorrelated along two independent axes (architecture family and input resolution), not from extra seeds; this reduces a two-backbone baseline from 0.354 to about 0.22–0.27 on a leakage-free captured proxy. For comparison, a public-leaderboard-optimised ensemble reaches 0.00041 on the public split but collapses to near chance on captured data, which is why we do not submit it. All training uses competition data and self-supervised augmentation only. Inference is a plain probability average with no test-time parameter fitting, packaged in a pinned container that runs fully offline.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026), Kaggle [the-freuid-challenge-2026-ijcai-ecai](https://kaggle.com/freuid-challenge-2026-ijcai-ecai) [7]

Kaggle team: ETRI_MobilityAI

Code: <https://github.com/hoppery/freuid-challenge-2026>

1 Introduction

Identity-document fraud has several threat models: physical manipulation and re-capture, localized digital editing, and fully synthetic (GenAI) content. The FREUID Challenge [7] combines

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these into a single task. The model outputs $P(\text{fraudulent}) \in [0, 1]$ for each image, scored by the FREUID Score (AuDET + APCER@1%BPCER; lower is better).

The challenge splits the test data in a deliberate way. The *public* split is about 5% of the test data. It is born-digital and covers the five training document types. The *private* split is about 95% and decides the full ranking. It is drawn from a hybrid of synthetic and physically captured images, and it adds **two document types that are not in training**. A detector that only memorises the born-digital appearance of the seen types generalises poorly to the captured and unseen-type content. The organizers also caution against optimising the public leaderboard [7].

A single submission is scored on both splits, and the private split decides the ranking. We cannot produce different predictions for the two splits, and per-image routing is unsafe (Section 5.5). We therefore submit one model. Section 5.2 shows why we do not submit the public-leaderboard winner: a model optimised for the born-digital split drops to near chance on captured data. Because the private split carries captured and unseen-type content, and because the metric punishes any subpopulation on which a model collapses, we submit a model built for robustness (our **private-best** model). We select it on a captured proxy, never on private labels.

Contributions.

1. We show that a model which tops the born-digital public leaderboard collapses on captured data, so public-leaderboard optimisation does not transfer to the private split that decides the ranking (Section 5.2).
2. We propose the Double-Trace Consistency detector (DTC), a detector whose patch-consistency signal does not depend on the document type (Section 2.2, Fig. 1).
3. We show a diversity-axis ensembling principle: decorrelation along architecture and resolution improves both tracks, while extra seeds do not (Sections 2.4, 5).

Related work. Manipulation and re-capture leave low-level traces that semantic features miss. High-pass residual streams expose splicing and reprint artifacts. Central Difference Convolutions capture the gradient micro-texture used in face anti-spoofing [12]. We build on the residual-trace idea, but we summarise it as a *consistency* statistic. Self-blended images [8] synthesise manipulation traces without manipulation labels. We adopt this idea for our self-supervised consistency target and for the field-tamper augmentation. We use self-supervised ViT features (DINOv2 [5], ViT [2]) together with convolutional families (ConvNeXt V2 [10], RegNet [6]) through `timm` [9]. These families have different inductive biases, which we use for decorrelation. For robustness we use Fourier Domain Adaptation [11] to randomise low-frequency appearance. Our ensembling builds on deep ensembles [3]; we add evidence on which type of diversity helps here.

2 Method

2.1 Overview and problem setting

Let x be an image and $y \in \{0, 1\}$ its fraud label. We learn $f_\theta(x) = \sigma(\ell_\theta(x)) \in [0, 1]$ and submit $f_\theta(x)$. Training uses the born-digital seen-type distribution $\mathcal{D}_{\text{tr}}$. Evaluation uses a public distribution $\mathcal{D}_{\text{pub}} \approx \mathcal{D}_{\text{tr}}$ and a private distribution $\mathcal{D}_{\text{prv}}$. The private distribution is shifted in two ways: acquisition (captured vs. born-digital) and label space (two unseen document types). The private distribution is not available, and it conflicts with $\mathcal{D}_{\text{pub}}$ (Section 5.2). We therefore do not train a single θ . We train one model per distribution. Every model uses the same feed-forward pipeline: decode $\rightarrow$ resize with ImageNet normalisation $\rightarrow$ DTC detector $\rightarrow$ sigmoid $\rightarrow$ probability average.

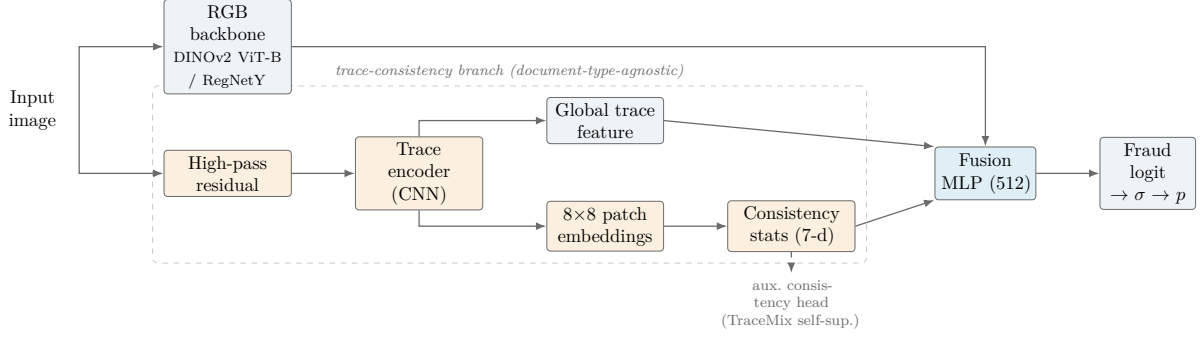


Figure 1: The Double-Trace Consistency detector (DTC). A pretrained RGB backbone is fused with a high-pass forensic branch. The patch-consistency statistics measure whether the acquisition trace is internally coherent. This signal does not depend on document-type appearance. Each deployed model is one such detector. The private-best model averages three of them (Section 2.4).

2.2 Double-Trace Consistency detector

Each detector (Fig. 1) combines an ImageNet/DINOv2-pretrained RGB backbone with a forensic branch. The forensic branch applies a Gaussian high-pass residual and a small strided CNN (**TraceEncoder**) that keeps the spatial layout. It produces (i) a global trace feature and (ii) an 8×8 grid of L2-normalised patch trace embeddings $\{z_i\}$. We take the off-diagonal pairwise cosine similarities of $\{z_i\}$ and compute a 7-dimensional **consistency statistic** $s(x)$ (mean, standard deviation, minimum, low percentiles, and per-patch max-similarity summaries). The name comes from the two uses of the high-pass trace: the global trace feature and the patch-level consistency signal $s(x)$. A genuine capture, including a legitimately re-captured one, has a uniform trace. A locally edited and reprinted forgery does not. The statistic $s(x)$ does not depend on the document type, so an unseen-type genuine document is not flagged only because it looks unfamiliar. The fraud logit fuses $[\text{RGB feature}; \text{global trace}; s(x)]$ through an MLP (Linear $\rightarrow$ 512 $\rightarrow$ GELU $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear $\rightarrow$ 1). An auxiliary head on $s(x)$ is trained on self-supervised **TraceMix** coherent/incoherent labels, which are independent of the fraud label. The branch does not depend on the backbone, so any RGB family can host it.

2.3 Capture robustness by data, not architecture

Robustness to captured documents comes from *data and augmentation*, not from architecture or loss changes (Section 5.5). The private-best models train on FREUID plus diverse real captured-genuine corpora (Section 3) with heavy print-and-capture augmentation and FDA domain randomisation [11]. With probability 0.4 we swap the low-frequency amplitude spectrum of an image with a donor of a *different* document type ($\beta \in [0.01, 0.08]$). This randomises the type appearance and keeps the forensic phase.

2.4 Diversity-axis ensembling

A single capture detector is limited. The gain comes from averaging backbones that are *decorrelated along independent axes*. We treat architecture family and input resolution as two separate diversity axes. Two ViT-B detectors at different resolutions are strongly correlated (Spearman $\rho \approx 0.79$). A pure-CNN family at a different resolution is much less correlated ($\rho \approx 0.5\text{--}0.7$). Adding such a member crosses both axes at once, and it gives the largest ensemble gain (Fig. 4). Extra *seeds* of the same architecture and resolution ($\rho > 0.99$) do not help. This principle applies to both tracks (Section 5).

2.5 The submitted model

We submit one model, the **private-best** ensemble: a probability average of three DTC detectors, selected on a captured proxy: ViT-B/14 DINOv2 at 896 and 1120px, and RegNetY-160 at 1120px. The two ViT-B anchors stabilise the operating point, because their score is stable across seeds. The decorrelated RegNetY adds the ranking gain. Across three seeds, this configuration had the smallest variance among the candidates. This is the model shipped in the reproducibility container.

For analysis we also build a **public-best** ensemble that we do *not* submit: a diversity-axis ensemble (ViT-L, ConvNeXt V2, ViT-Giant across resolutions) with a *field-tamper* augmentation that targets born-digital tampering. It tops the public leaderboard but collapses on captured data (Section 5.2), so it is a reference, not our submission.

2.6 Score calibration

Each detector outputs a fraud logit. A sigmoid maps it to a per-model probability. The submission score is the arithmetic mean of the ensemble probabilities. We apply no temperature or learned calibration, and no post-processing differs between train and test time. Unreadable images are assigned 0.5, so every input gives exactly one finite score in $[0, 1]$.

3 Data

We use the official FREUID `train/` images and `train.csv` metadata (Kaggle competition **the-freuid-challenge-2026-ijcai-ecai**). The private-best models also train on diverse *captured-genuine* corpora (moderate upweight, no external fraud labels): MIDV-500 [1], MIDV-2019, MIDV-Holo, MIDV-2020, and BID (Brazilian CNH/RG/CPF photographs). We combine these with FREUID into a single manifest. FantasyID, a public research dataset of fantasy ID cards (Idiap), is held out as a leakage-free *captured proxy*. It is a separate corpus, so we never train on it; we use it only for model selection. Its composition is described in the Validation paragraph below. All corpora are used under their public research licenses, and the pretrained weights under their upstream licenses.

Validation. The public leaderboard reflects the born-digital distribution, so it does not predict the captured private set (Section 5.2). We therefore select the private-best model on a captured proxy, the FantasyID test split: 1,295 images across six document styles (465 genuine cards that are physically printed and photographed, and 830 GenAI-manipulated versions with face-swap or text edits). We chose FantasyID because it matches the two stated properties of the private test (Section 1): its genuine cards are captured rather than born-digital, and its document styles differ from the five FREUID training types. We do not claim that FantasyID and the private set share the same distribution, and FantasyID is not an organizer-recommended proxy; its attack types in particular may differ from the private test. We therefore use it only for model selection and read its FREUID score as an indicative lower bound. We hold out a random 10% split of the training manifest only for logging. The private set is not available at submission time, so proxy selection is the only option.

4 Inference

The container entrypoint indexes a flat directory of images. It runs the three checkpoints (build with `pretrained=False`, load embedded weights, forward pass, sigmoid) and writes the per-image probability average as `id,label1`. Test-time preprocessing is a plain resize to the native resolution with ImageNet normalisation. The deployed models use no tiling, no multi-crop, and

Table 1: Main results. FREUID Score $\downarrow$ (lower is better). We submit the private-best model; the public-best is a public-leaderboard reference we do not submit, because it collapses on captured data (Fig. 2). Captured-proxy figures are a small cross-corpus lower bound.

Model	Public LB $\downarrow$	Captured proxy $\downarrow$
Private-best (submitted)	0.169	$\approx 0.22\text{--}0.27$
Public-best (reference, not submitted)	0.00041	≈ 0.99 (chance)

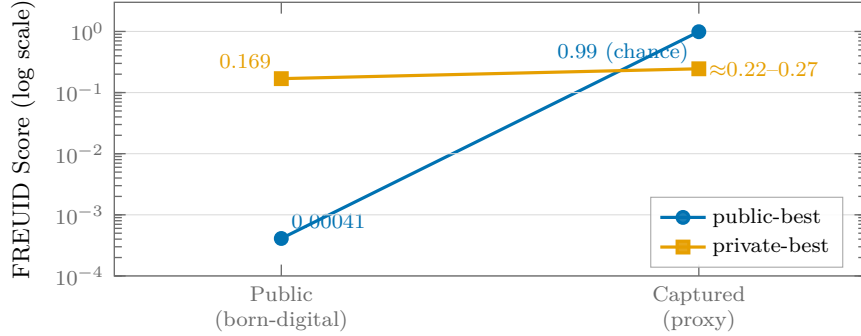


Figure 2: Each model fits only one distribution. A model is strong (low) on its own distribution and weak (high) on the other, so the two lines cross. Lower is better.

no test-time augmentation. The device is CUDA if available, else CPU. The per-model batch size follows the training config. Unreadable images are assigned 0.5. The exact command is in Section 6.

5 Results

5.1 Main results

Table 1 reports the submitted private-best model and the public-best reference on both splits. The private-best model reduces a two-backbone baseline by about half on the captured proxy, and it stays well above chance on born-digital data. The public-best reference reaches a public FREUID Score of 0.00041 (with three models; 0.00039 for the full ensemble), but it collapses on captured data, so we do not submit it.

5.2 Each model works on only one distribution

Fig. 2 plots each model on both distributions. The two lines cross. The public-best model is strong on born-digital data (0.00041) but at chance on captured data (about 0.99; a random classifier scores about 0.99 on APCER@1%BPCER). The reason is that the public-best model relies on born-digital forensic cues, such as fine editing traces and JPEG discontinuities. Print-and-capture re-acquisition overwrites these cues with a uniform capture signature, so they no longer separate genuine from fraud. The private-best model shows the opposite order. The gap between the two models is about $400\times$ on the public set, and the order fully reverses on the captured proxy. A single model therefore cannot serve both distributions, and the public score does not predict private performance. The private-best model stays well above chance on born-digital data (0.169), so it is our submitted model: it does not collapse on either split, and it is the stronger choice on the captured and unseen-type content that the private split contains and the public split does not.

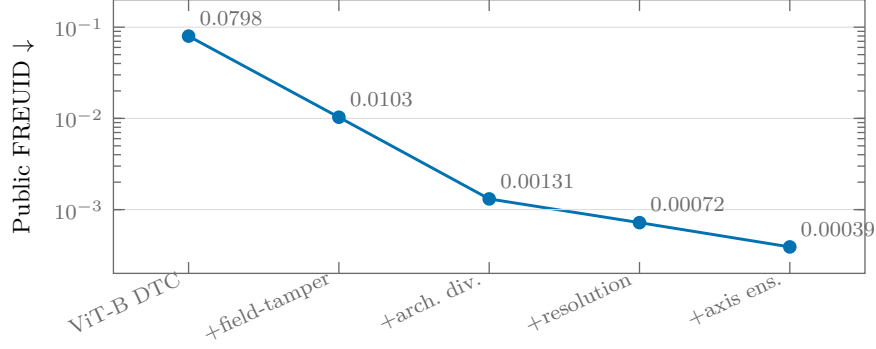


Figure 3: The diversity-axis principle on the public-best reference. Each added component lowers the public FREUID Score. The full ensemble reaches 0.00039; the three-model version scores 0.00041. This model is not submitted.

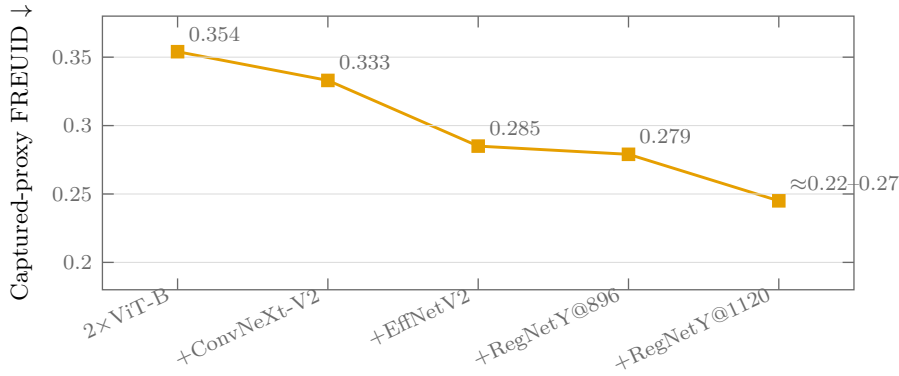


Figure 4: Ensemble ladder for the submitted model. Adding decorrelated architecture/resolution members lowers the captured-proxy FREUID Score. The last point is a range. The plotted value is its midpoint.

5.3 Diversity axes beat seeds (measured on the public leaderboard)

The diversity-axis principle is easiest to measure on the public leaderboard, where the labels are real. Fig. 3 traces the public FREUID Score as we add each component to the public-best reference. The field-tamper augmentation gives the first large drop. Adding architecture diversity, then resolution diversity, and then an axis-crossing member each lowers the score further. Seed replicas (not shown) do not change the score. We apply the same principle to the submitted model (Section 5.4).

5.4 Ensemble ladder (submitted model)

Fig. 4 shows the same principle on the submitted model, measured on the captured proxy. We start from two correlated ViT-B detectors. Each new decorrelated family lowers the proxy FREUID Score. The axis-crossing member (RegNetY-160 at 1120 px, a new family and a new resolution) gives the final step and reduces the two-ViT baseline by about half.

5.5 Approaches evaluated and rejected

We also tested approaches that did not help, and we dropped them. Submitting the public-best ensemble on captured data was capture-blind (about chance). Adding MIDV-2020 to the capture mix gave no gain, because of a distribution mismatch. A learned domain router keyed on framing and context, and it misrouted cropped captured images. Warm-starting the capture model from the public-best checkpoint gave a worse peak, because the fraud-specific features

do not survive re-capture. Architecture and loss levers on the capture axis (focal loss [4], CDC convolutions [12], MixStyle [13]) did worse than the data-diversity recipe.

6 Reproducibility

The reproducibility container ships the private-best model. The weights are baked into the image, and the models build with `pretrained=False`. The container therefore runs with `-network none` (no runtime downloads) and writes only to `/submissions`.

- **Environment (pinned):** `torch==2.12.0` (CUDA 13), `timm==1.0.27`, `albumentations==2.0.8`, `opencv-python-headless==4.13.0`, `numpy==2.4.6`, `pandas==3.0.3`, `scipy==1.17.1`, `scikit-learn==1.9.0`, `pillow==12.2.0`, `pyyaml==6.0.2`, `pyarrow`; Python ≥ 3.11 .
- **Hardware / runtime:** training on 2× NVIDIA RTX PRO 6000 (Blackwell, 96 GB); inference uses a single GPU. The CPU fallback works but is slow.
- **Randomness:** global seed 42. CUDA/cuDNN and AMP add minor non-determinism, within the proxy’s seed noise.
- **Repository:** <https://github.com/hopper/freuid-challenge-2026> (MIT license; the frozen commit SHA is in our reproducibility reply).

Build and run (organizer contract):

```
docker build -t freuid-repro:local .
docker run --network none \
  -v <flat_images>:/data:ro \
  -v "$(pwd)/out":/submissions \
  freuid-repro:local          # -> out/submission.csv (id,label)
```

7 Team and contributions

Both authors contributed to the method, the experiments, and the manuscript. Su-Yong An (corresponding author) led model development, training, and evaluation. Lae-Kyoung Lee contributed to the methodology, the analysis, and the manuscript.

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A Hyperparameters

Table 2: Per-model hyperparameters of the private-best ensemble. Shared: AdamW, weight decay 0.05, cosine LR schedule, 3 epochs, seed 42, AMP, DTC consistency weight $\lambda_{\text{dte}}=0.5$, `tracemix_p=0.5`, `fda_p=0.4`, heavy print-and-capture augmentation.

Hyperparameter	ViT-B @896	ViT-B @1120	RegNetY-160 @1120
Backbone	<code>vit_base_patch14_reg4_dinov2</code>		<code>regnety_160.tv2_in1k</code>
Image size	896	1120	1120
Batch size	24	12	8
Learning rate	1×10^{-4}	1×10^{-4}	5×10^{-5}
Weight decay	0.05	0.05	0.05
Epoch used	2	2	1
Backbone freezing	last 2 blocks + norm	last 2 blocks + norm	full fine-tune
Seed	42	42	42

B Additional ablations

See Figs. 3 and 4 (diversity-axis contributions on each track) and the rejected-approaches list in Section 5.5. Further ablations (capture-source diversity and resolution sweeps) are available in the repository notes.